

Documentation

Introduction:

This is the documentation of Legal Case Analyser Python Automation Script.

This script is divided into 4 parts

1. Web Scrapers of legal articles and legal proceedings.

- Baili web scrapper.
- Civil litigation brief scraper.
- Cost Barrister scraper.
- Law Gazette scraper.
- LinkedIn posts scraper.

2. Summarisers for legal articles and legal proceedings.

- Baili case summarizer.
- Civil litigation brief summarizer.
- Cost Barrister summarizer.
- Law Gazette summarizer.

3. Automatic mail sender of summarized texts.

4. Automatic upload of summarized texts to Google Drive.

If we look at the total there are 11 Python scripts for this project.

System Requirements:

- Ram: 8 GB RAM.
- Free Space in disc: at least 15 GB (to install hugging face models into the local machine).
- Python version 3 or above installed.
- OS: Windows/Mac/Linux

Python libraries and modules are required, as well as their installation:

- Modules and libraries
 1. os
 2. nltk
 3. spacy
 4. RecursiveCharacterTextSplitter
 5. AutoModelForQuestionAnswering
 6. AutoTokenizer
 7. AutoModelForSeq2SeqLM
 8. PegasusTokenizer
 9. PegasusForConditionalGeneration
 10. pipeline
 11. transformers
 12. pandas
 13. re
 14. language_tool_python
 15. requests
 16. BeautifulSoup
 17. csv
 18. feedparser
 19. newspaper
 20. Article
 21. Selenium
 22. smtplib

23. Google OAuth 2.0

- Installation:

If you're on macOS, you might need to use `pip3` instead of `pip` if Python 3 is installed as `python3` else, if you are using Windows then use the `pip` command and `python`.

All these libraries/modules should be installed after downloading Python.

1. `os` : This is a built-in library in Python, so no installation is needed.
2. `nltk` :

```
#WINDOWS
pip install nltk
#MAC/LINUX
pip3 install nltk
```

3. `spacy` :

```
#WINDOWS
pip install spacy
python -m spacy download en_core_web_sm
#if you are using MAC/LINUX then use:
pip3 install spacy
python3 -m spacy download en_core_web_sm
```

4. `RecursiveCharacterTextSplitter` : This is part of the `langchain` library:

```
#WINDOWS
pip install langchain
#MAC/LINUX
pip3 install langchain
```

5. `AutoModelForQuestionAnswering` :
6. `toTokenizer` :
7. `AutoTokenizer` :

8. `AutoModelForSeq2SeqLM`:
9. `PegasusTokenizer`:
10. `PegasusForConditionalGeneration`:
11. `pipeline`:
12. `transformers`:

```
#WINDOWS
pip install transformers
#MAC/LINUX
pip3 install transformers
```

13. `pandas`:

```
#WINDOWS
pip install pandas
#MAC/LINUX
pip3 install pandas
```

14. `re`: This is a built-in library in Python, so no installation is needed.

15. `language_tool_python`:

```
#WINODOWS
pip install language-tool-python
#MAC/LINUX
pip3 install language-tool-python
```

16. `requests`:

```
#WINDOWS
pip install requests
#MAC/LINUX
pip3 install requests
```

17. `BeautifulSoup`: The library is named `beautifulsoup4`.

```
#WINDOWS
pip install beautifulsoup4
#MAC/LINUX
pip3 install beautifulsoup4
```

18. `csv` : This is a built-in library in Python, so no installation is needed.

19. `feedparser` :

```
#WINDOWS
pip install feedparser
#MAC/LINUX
pip3 install feedparser
```

20. `newspaper` : The package is named `newspaper3k` .

```
#WINDOWS
pip install newspaper3k
pip install lxml Pillow
#MAC/LINUX
pip3 install newspaper3k
pip3 install lxml Pillow
```

21. `Article` : This is a part of the `newspaper3k` library.

22. `Selenium` :

```
#WINDOWS
pip install selenium
#MAC/LINUX
pip3 install selenium
```

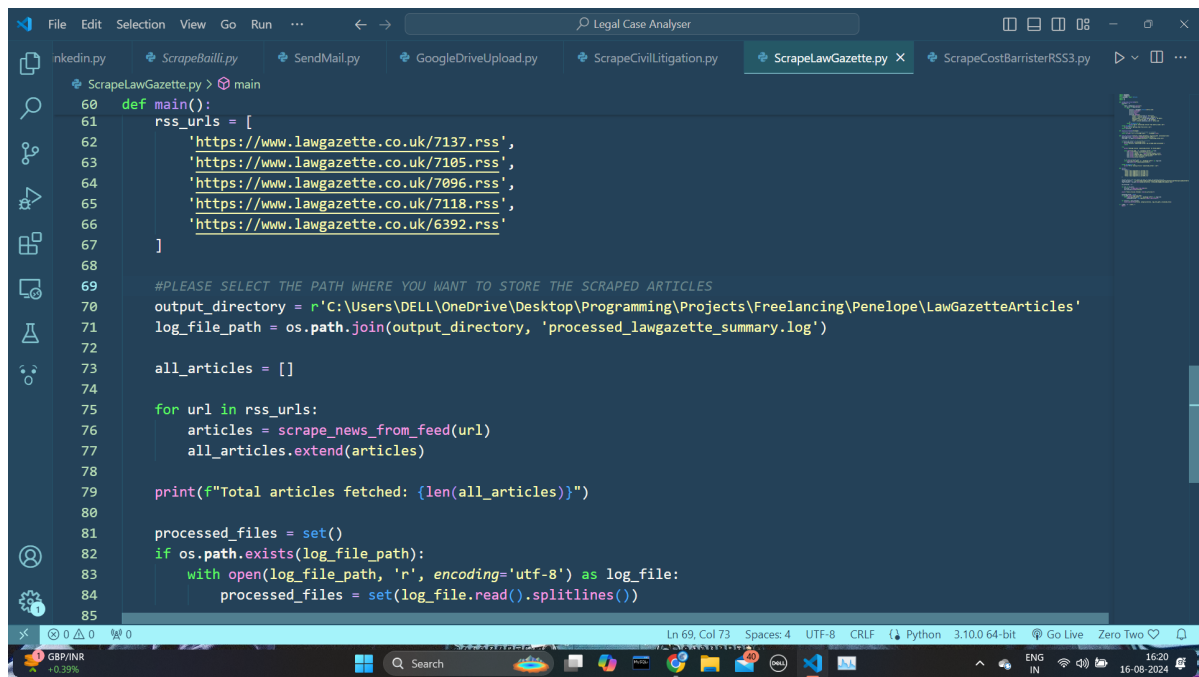
23. `smtplib` : This is a built-in library in Python, so no installation is needed.

24. **Google OAuth 2.0**: For OAuth 2.0, you generally use `google-auth` and `google-auth-oauthlib` :

```
#WINDOWS
pip install google-auth google-auth-oauthlib
#MAC/LINUX
pip3 install google-auth google-auth-oauthlib
```

Instructions to run the Python Web Scrapers:

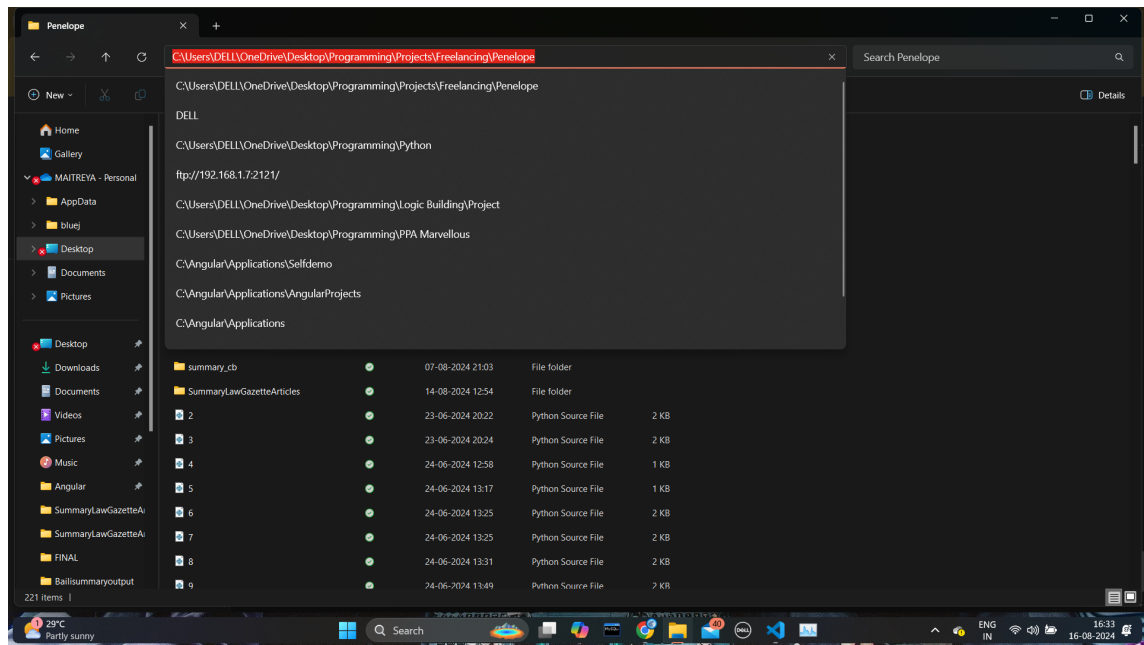
1. Before running the script we have to ensure where we have to store the scraped articles so we need to initialize the path manually in the code.
2. For example in ScrapeLawGazette.py program at line no 70 there's a variable storing the path where the articles should be stored. So just change it accordingly and save the program by pressing ctrl+S key.
3. Here's the example. Replace the path which is inside (' ') inverted commas.

A screenshot of a code editor window showing a Python script named 'ScrapeLawGazette.py'. The script is part of a project called 'Legal Case Analyser'. The code defines a 'main' function that sets up a list of RSS URLs from the Law Gazette website. It then defines an output directory and a log file path. The script iterates through the URLs, scrapes news from each feed, and prints the total number of articles fetched. It also checks if a log file exists and updates it with the current state. The status bar at the bottom indicates the file is at line 69, column 73, with 4 spaces, UTF-8 encoding, and CRLF line endings. The Python version is 3.10.0 64-bit. The system tray shows the date as 16-08-2024 and time as 16:20.

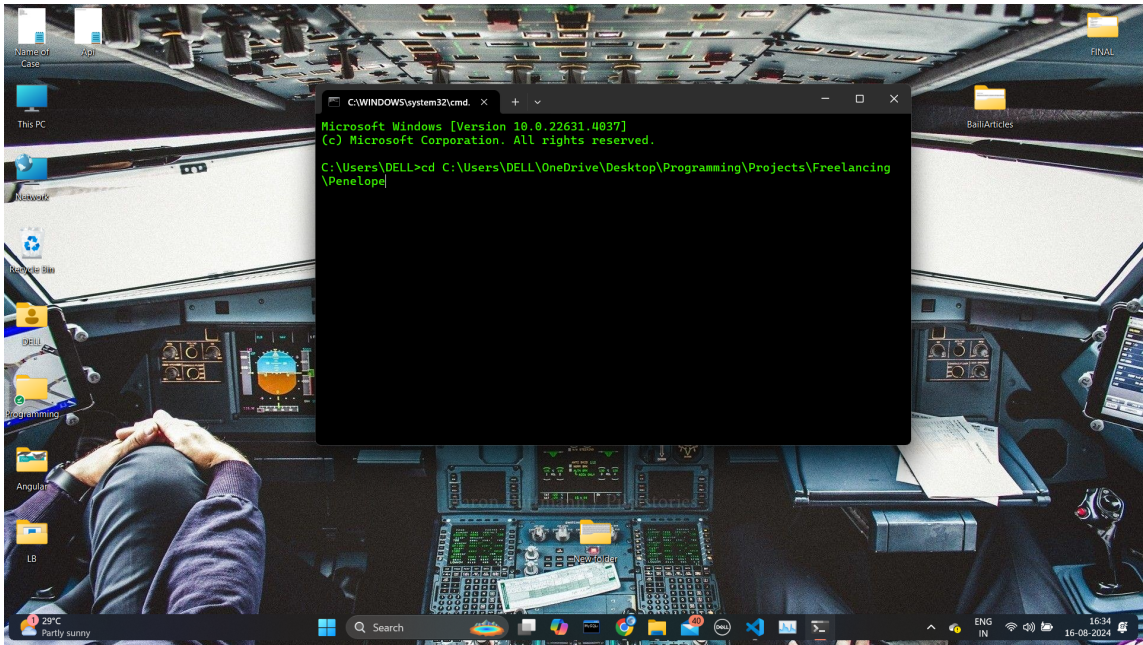
```
File Edit Selection View Go Run ... Legal Case Analyser
ScrapeLawGazette.py x
ScrapeBailli.py SendMail.py GoogleDriveUpload.py ScrapeCivilLitigation.py ScrapeLawGazette.py ScrapeCostBarristerRSS3.py
60 def main():
61     rss_urls = [
62         'https://www.lawgazette.co.uk/7137.rss',
63         'https://www.lawgazette.co.uk/7105.rss',
64         'https://www.lawgazette.co.uk/7096.rss',
65         'https://www.lawgazette.co.uk/7118.rss',
66         'https://www.lawgazette.co.uk/6392.rss'
67     ]
68
69     #PLEASE SELECT THE PATH WHERE YOU WANT TO STORE THE SCRAPED ARTICLES
70     output_directory = r'C:\Users\DELL\OneDrive\Desktop\Programming\Projects\Freelancing\Penelope\LawGazetteArticles'
71     log_file_path = os.path.join(output_directory, 'processed_lawgazette_summary.log')
72
73     all_articles = []
74
75     for url in rss_urls:
76         articles = scrape_news_from_feed(url)
77         all_articles.extend(articles)
78
79     print(f"Total articles fetched: {len(all_articles)}")
80
81     processed_files = set()
82     if os.path.exists(log_file_path):
83         with open(log_file_path, 'r', encoding='utf-8') as log_file:
84             processed_files = set(log_file.read().splitlines())
85
Ln 69, Col 73 Spaces: 4 UTF-8 CRLF Python 3.10.0 64-bit Go Live Zero Two
GBP/INR +0.39%
```

4. Then run the program.
5. For running this program follow the below steps:
 - a. Open command prompt or terminal

- b. Now select the path where you've stored on the device.
- c. For example : The ScrapeLawGazette.py file is present on this path



- d. Copy that path and paste it into terminal.
- e. Use `cd` command before path.
- f. eg : `cd`
`C:\Users\DELL\OneDrive\Desktop\Programming\Projects\Freelancing\Penelope`
- g. PRESS ENTER



h. If you're on windows platform run the program by typing

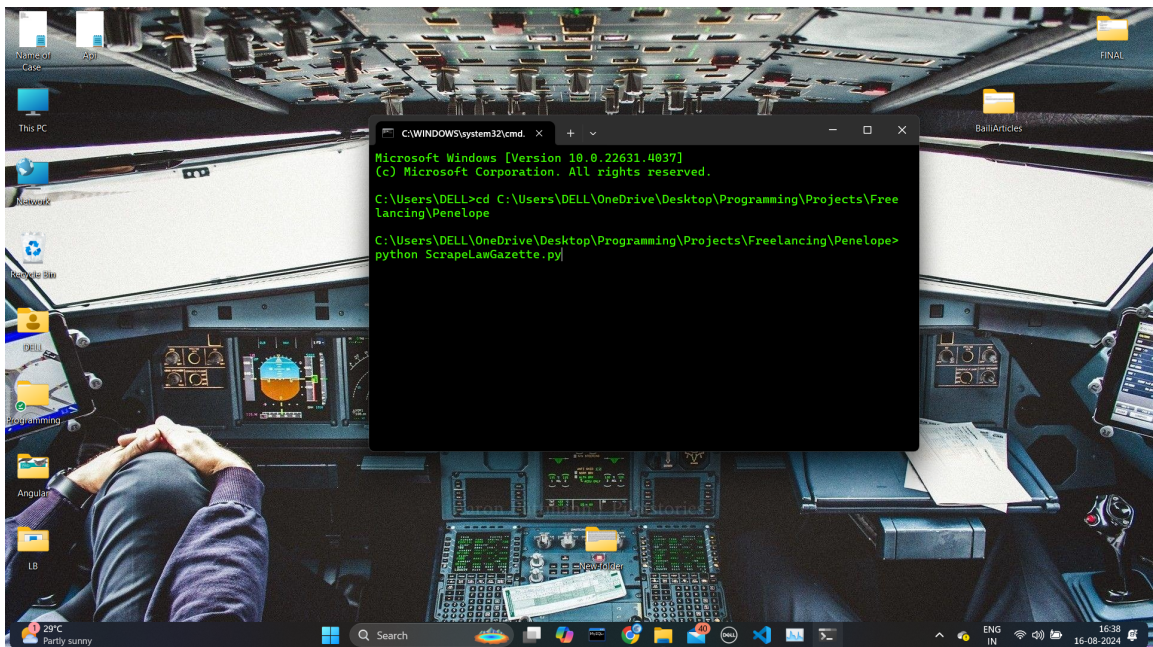
python ScrapeLawGazette.py

if you're on MAC/LINUX platform then run the program by typing

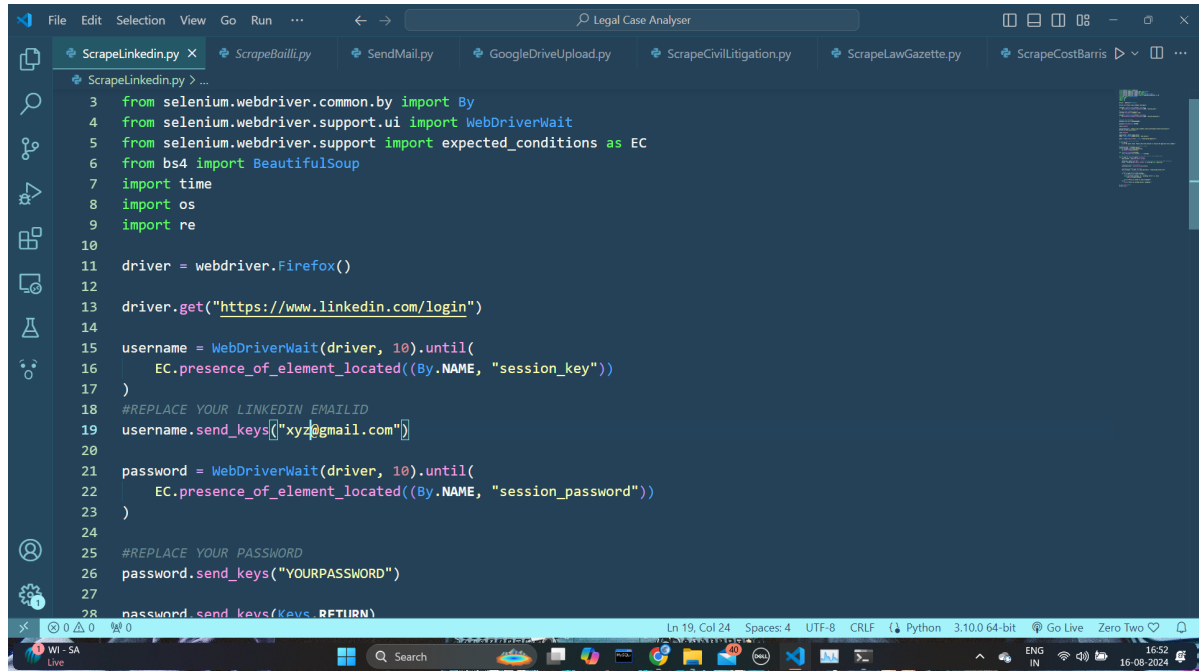
python3 ScrapeLawGazette.py

then PRESS ENTER

then the program will work and it will do the work it has assigned to.



6. The same instructions are as it is applicable to Summariser python scripts.
7. LinkedIn webscraper instructions:
 - a. PLEASE REPLACE YOUR EMAILID AND PASSWORD in the code at line 19 and line 26.

A screenshot of a code editor window titled 'Legal Case Analyser'. The editor shows a Python script named 'ScrapeLinkedIn.py'. The script uses Selenium and BeautifulSoup to automate a LinkedIn login. It imports necessary modules, initializes a Firefox driver, navigates to the LinkedIn login page, and waits for the session key and password fields. Lines 19 and 26 contain placeholder text for email and password, respectively, with comments instructing the user to replace them. The script ends with a return statement for the password field.

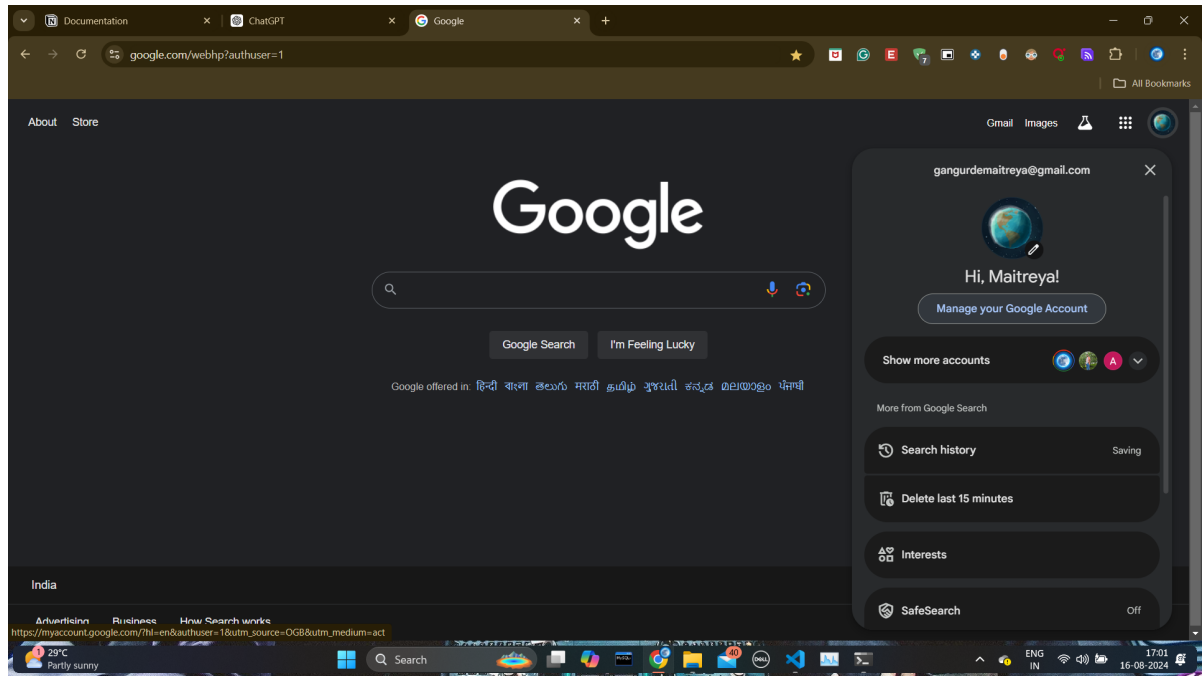
```
3 from selenium.webdriver.common.by import By
4 from selenium.webdriver.support.ui import WebDriverWait
5 from selenium.webdriver.support import expected_conditions as EC
6 from bs4 import BeautifulSoup
7 import time
8 import os
9 import re
10
11 driver = webdriver.Firefox()
12
13 driver.get("https://www.linkedin.com/login")
14
15 username = WebDriverWait(driver, 10).until(
16     EC.presence_of_element_located((By.NAME, "session_key"))
17 )
18 #REPLACE YOUR LINKEDIN EMAILID
19 username.send_keys("xyz@gmail.com")
20
21 password = WebDriverWait(driver, 10).until(
22     EC.presence_of_element_located((By.NAME, "session_password"))
23 )
24
25 #REPLACE YOUR PASSWORD
26 password.send_keys("YOURPASSWORD")
27
28 password.send_keys(Keys.RETURN)
```

Instructions to run Python summarizers:

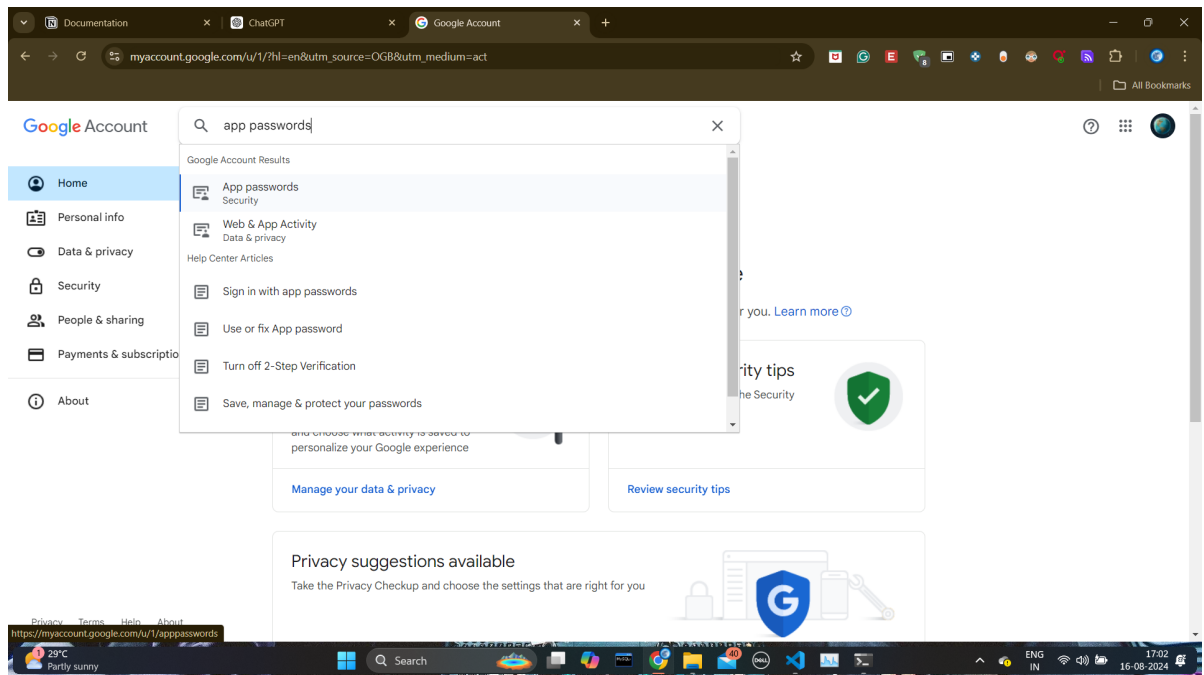
1. Follow the path instructions as specified above.
2. Run the program as specified above.

Instructions to run Email Sender script.

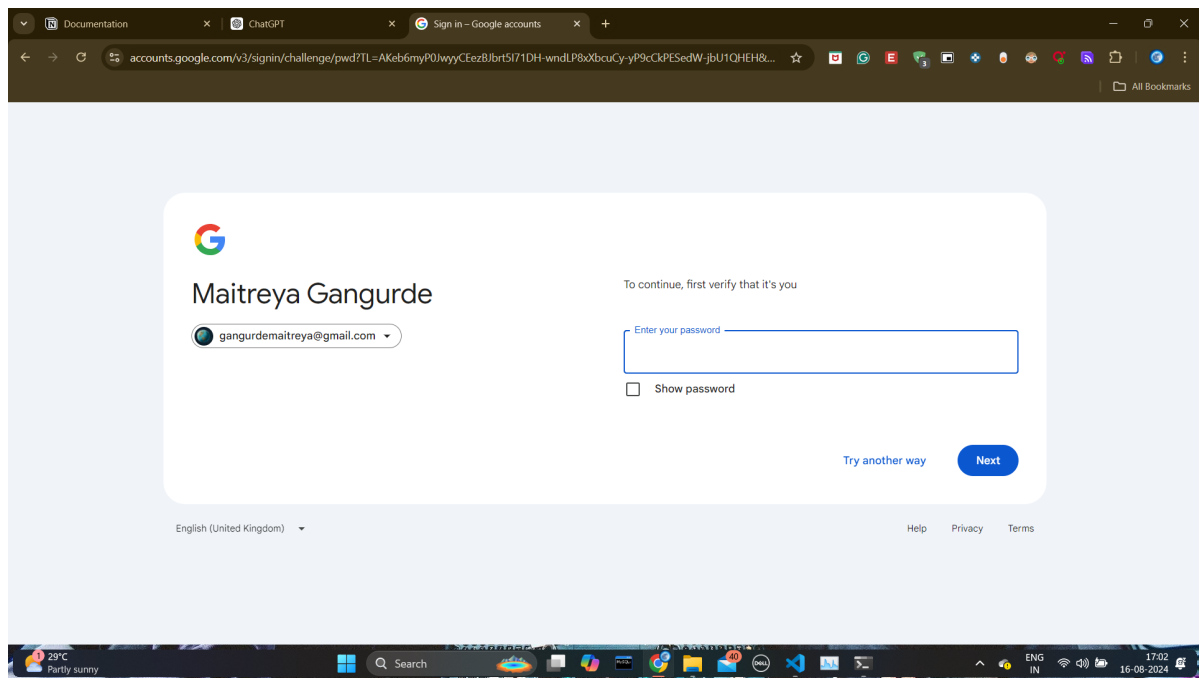
1. Before running the script successfully we need to first get app passwords from google.
2. Follow the steps to fetch app passwords
 - a. Go to manage your manage accounts sections



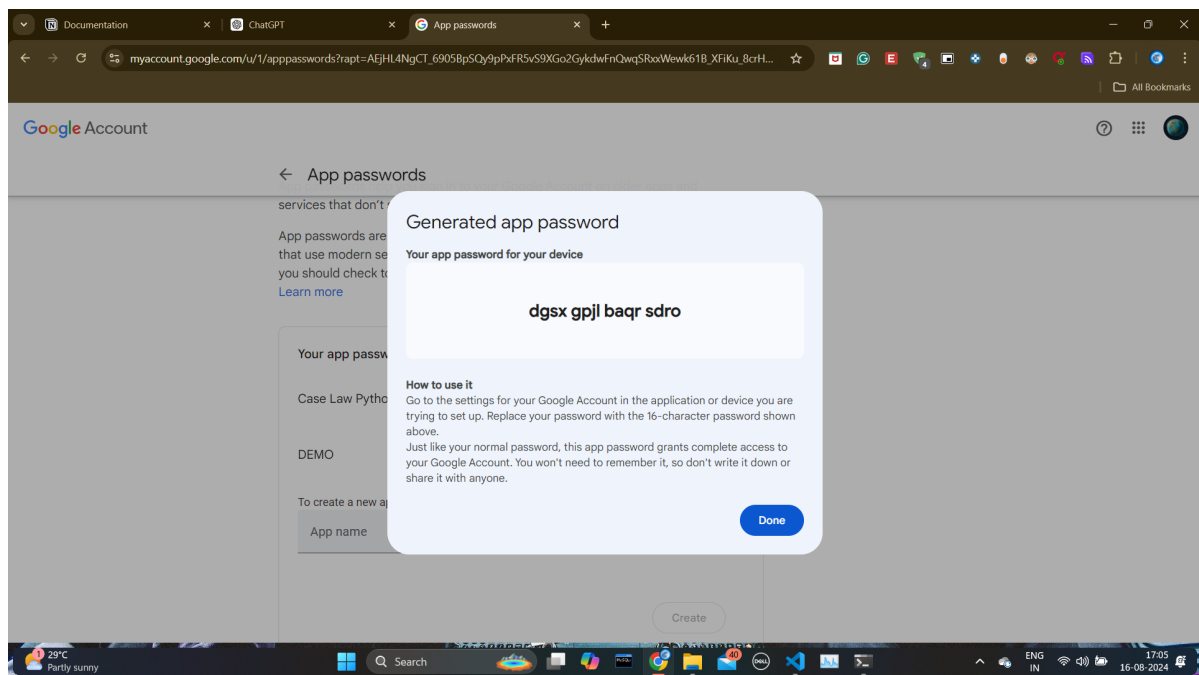
b. Then type app passwords



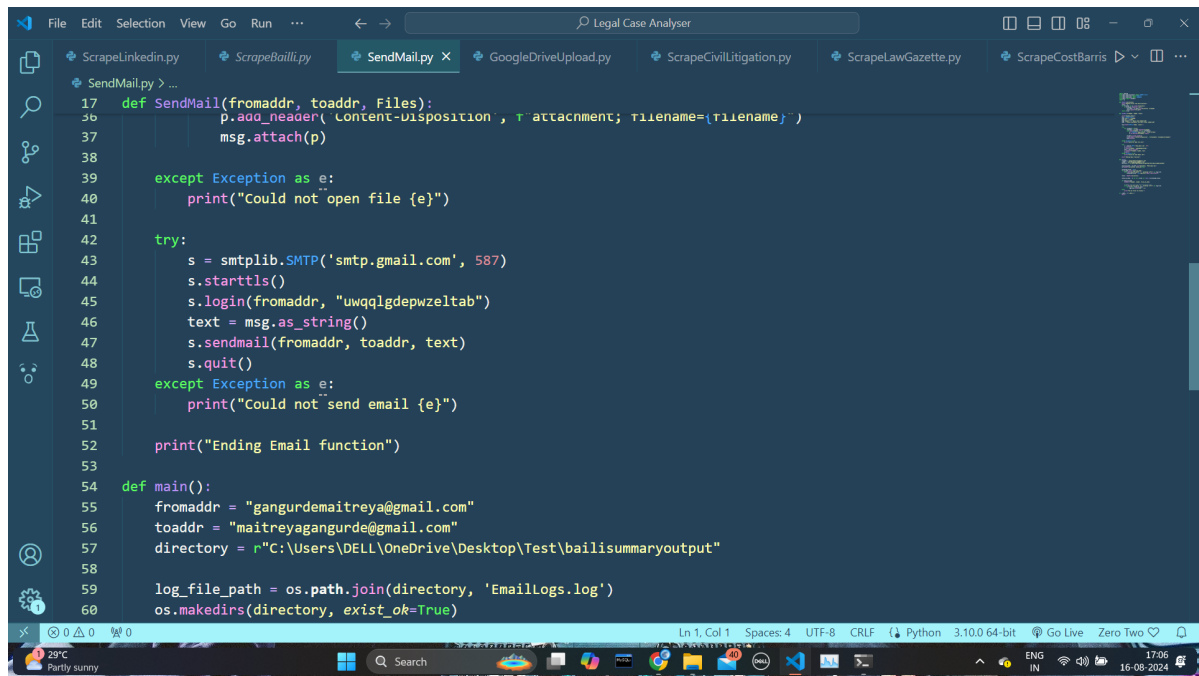
c. Select app passwords and login



d. Create app name and press create then immediately copy the password as it is one time display.



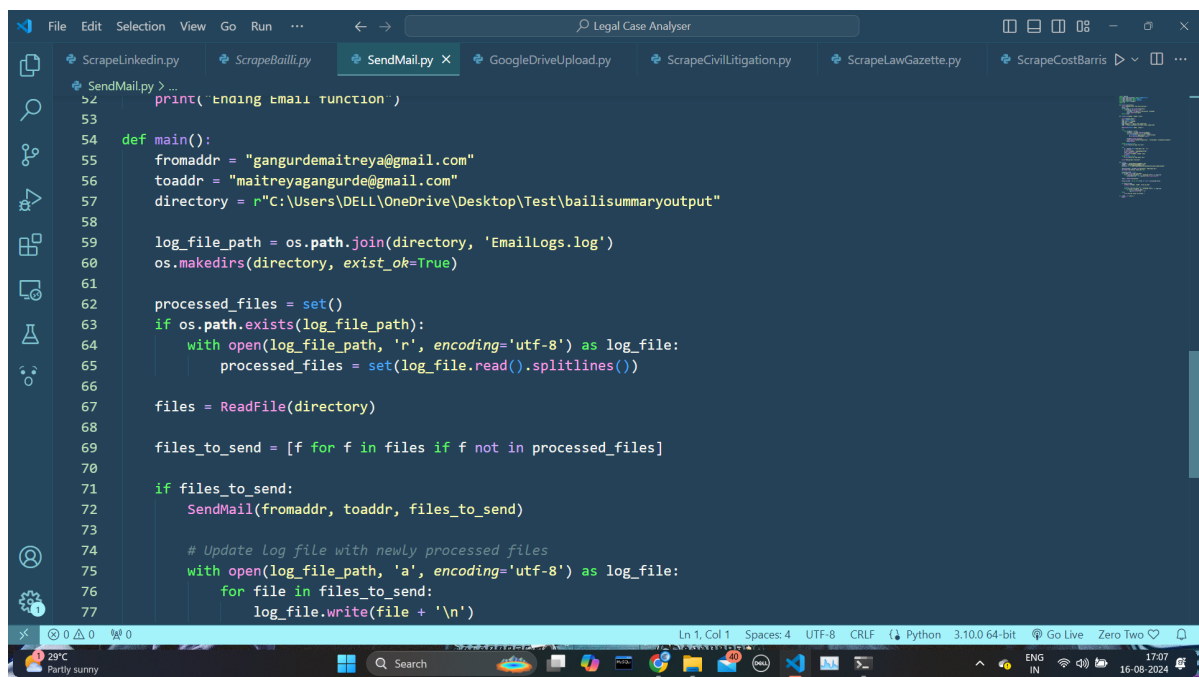
e. Paste this password in the program.



```
17 def SendMail(fromaddr, toaddr, Files):
18     p.add_header('Content-Disposition', 'attachment; filename={filename}')
19     msg.attach(p)
20
21     except Exception as e:
22         print("Could not open file {e}")
23
24     try:
25         s = smtplib.SMTP('smtp.gmail.com', 587)
26         s.starttls()
27         s.login(fromaddr, "uwqlgdepwzeltab")
28         text = msg.as_string()
29         s.sendmail(fromaddr, toaddr, text)
30         s.quit()
31     except Exception as e:
32         print("Could not send email {e}")
33
34     print("Ending Email function")
35
36 def main():
37     fromaddr = "gangurdemaitreya@gmail.com"
38     toaddr = "maitreyagangurde@gmail.com"
39     directory = r"C:\Users\DELL\OneDrive\Desktop\Test\bailisummaryoutput"
40
41     log_file_path = os.path.join(directory, 'EmailLogs.log')
42     os.makedirs(directory, exist_ok=True)
```

see line no 45. Paste the password.

f. Enter Sender address and receiver address and also select the directory where the summarised file exists.

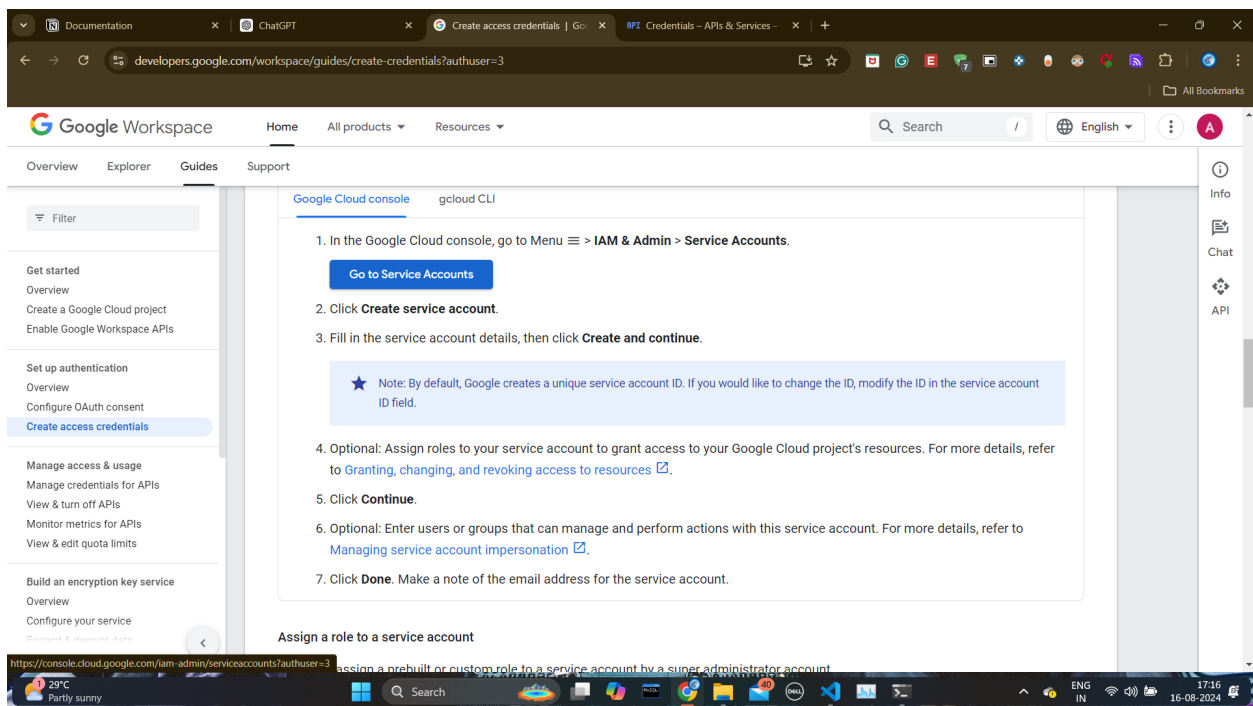


```
53     print("Ending Email function")
54
55     def main():
56         fromaddr = "gangurdemaitreya@gmail.com"
57         toaddr = "maitreyagangurde@gmail.com"
58         directory = r"C:\Users\DELL\OneDrive\Desktop\Test\bailisummaryoutput"
59
60         log_file_path = os.path.join(directory, 'EmailLogs.log')
61         os.makedirs(directory, exist_ok=True)
62
63         processed_files = set()
64         if os.path.exists(log_file_path):
65             with open(log_file_path, 'r', encoding='utf-8') as log_file:
66                 processed_files = set(log_file.read().splitlines())
67
68         files = ReadFile(directory)
69
70         files_to_send = [f for f in files if f not in processed_files]
71
72         if files_to_send:
73             SendMail(fromaddr, toaddr, files_to_send)
74
75             # Update log file with newly processed files
76             with open(log_file_path, 'a', encoding='utf-8') as log_file:
77                 for file in files_to_send:
78                     log_file.write(file + '\n')
```

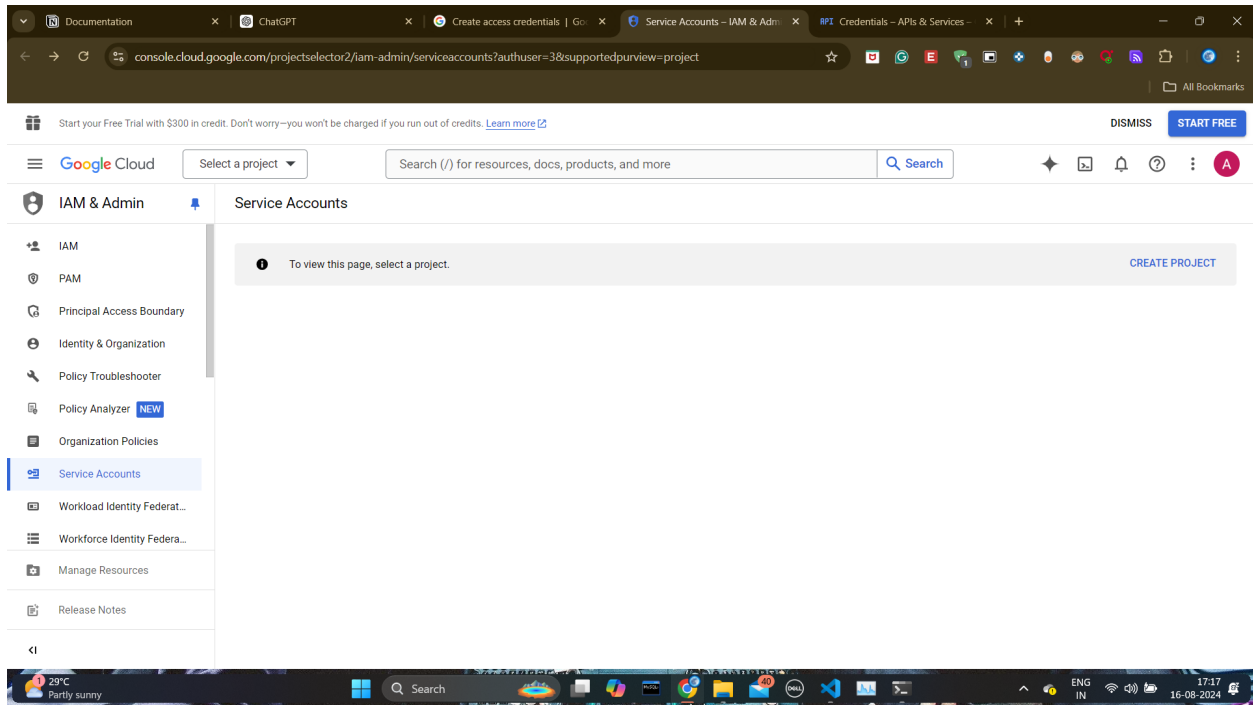
g. Run the program.

Instructions to run the File Upload Program:

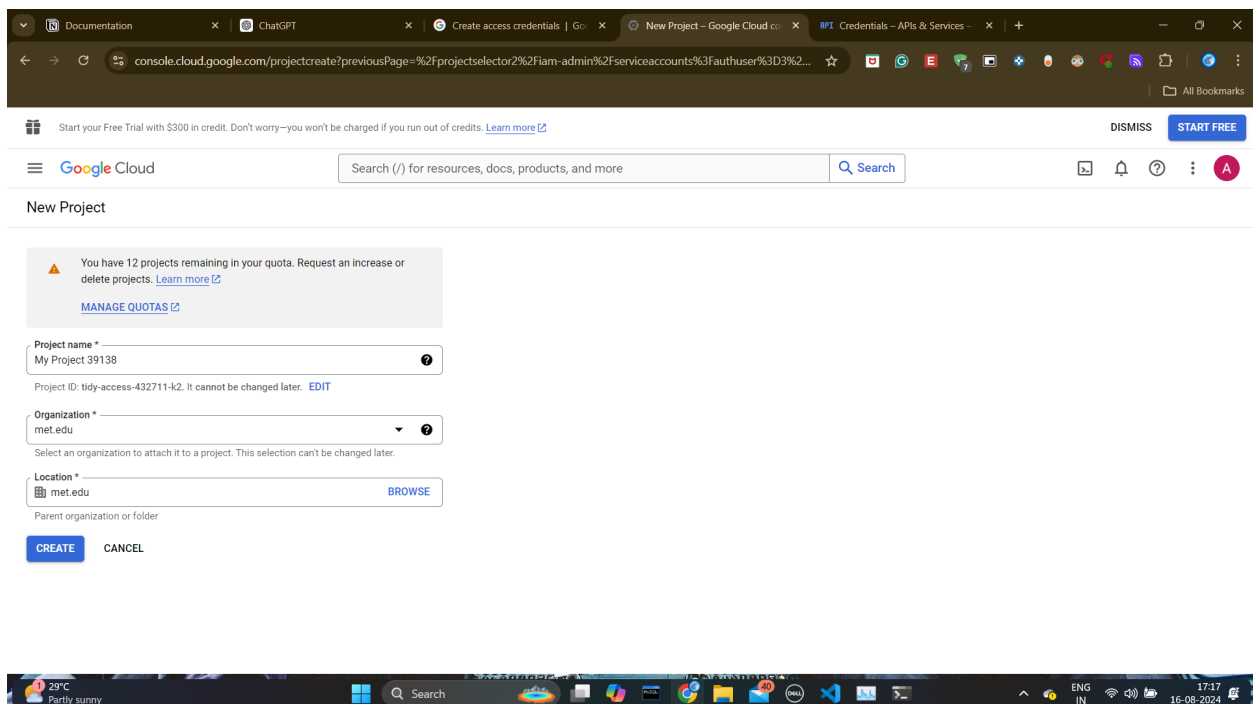
1. We are using GOOGLE DRIVE API
2. search **create access credentials** on google.
<https://developers.google.com/workspace/guides/create-credentials>
3. Then go to service accounts.



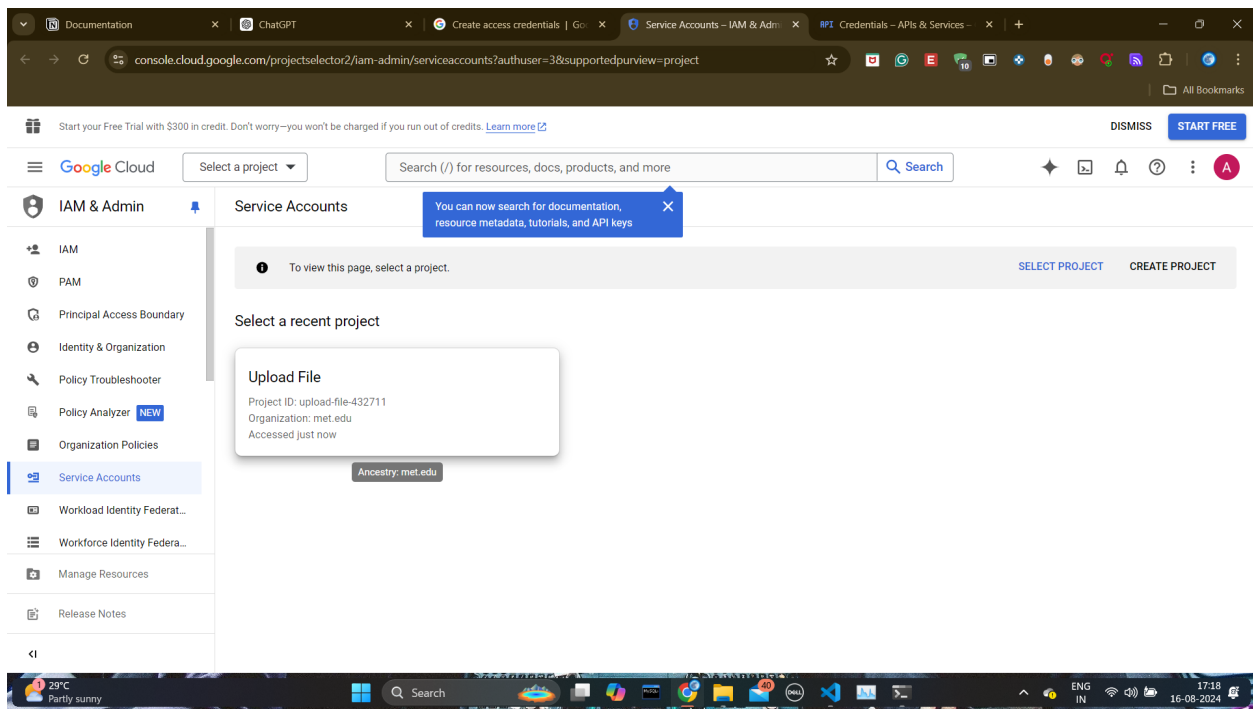
4. Press create project



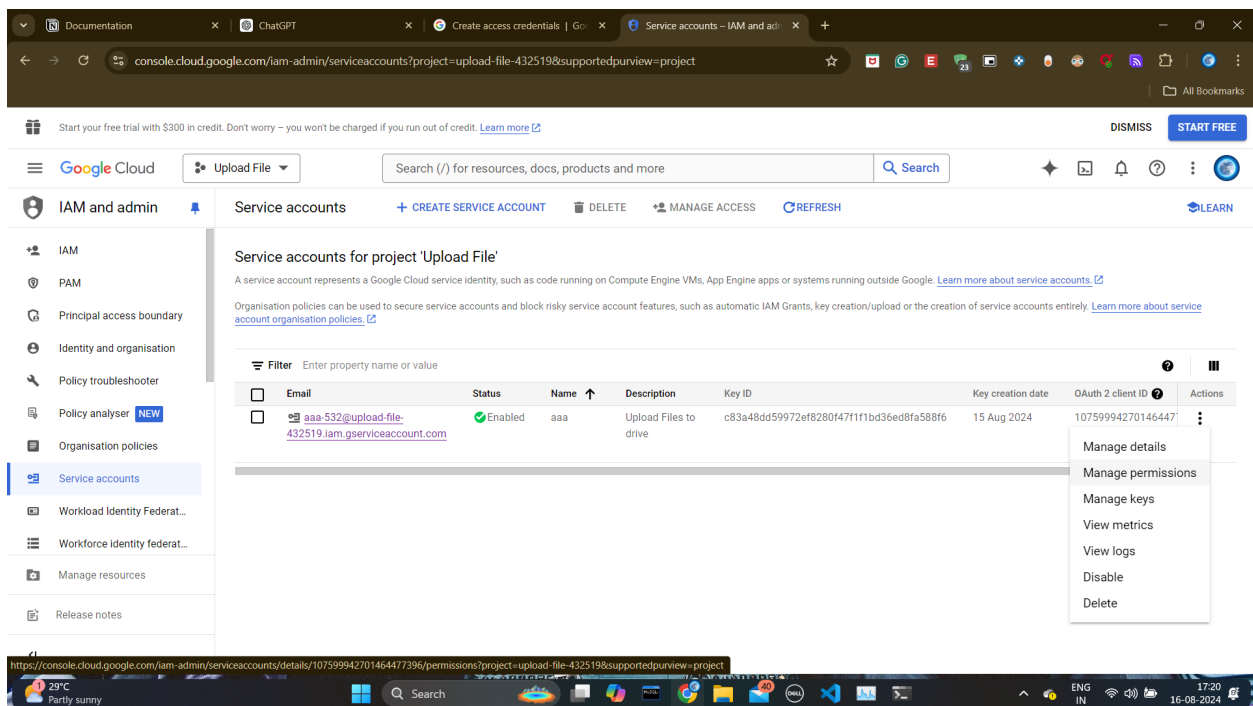
5. Enter your project name etc



6. Select the project that we created just now.



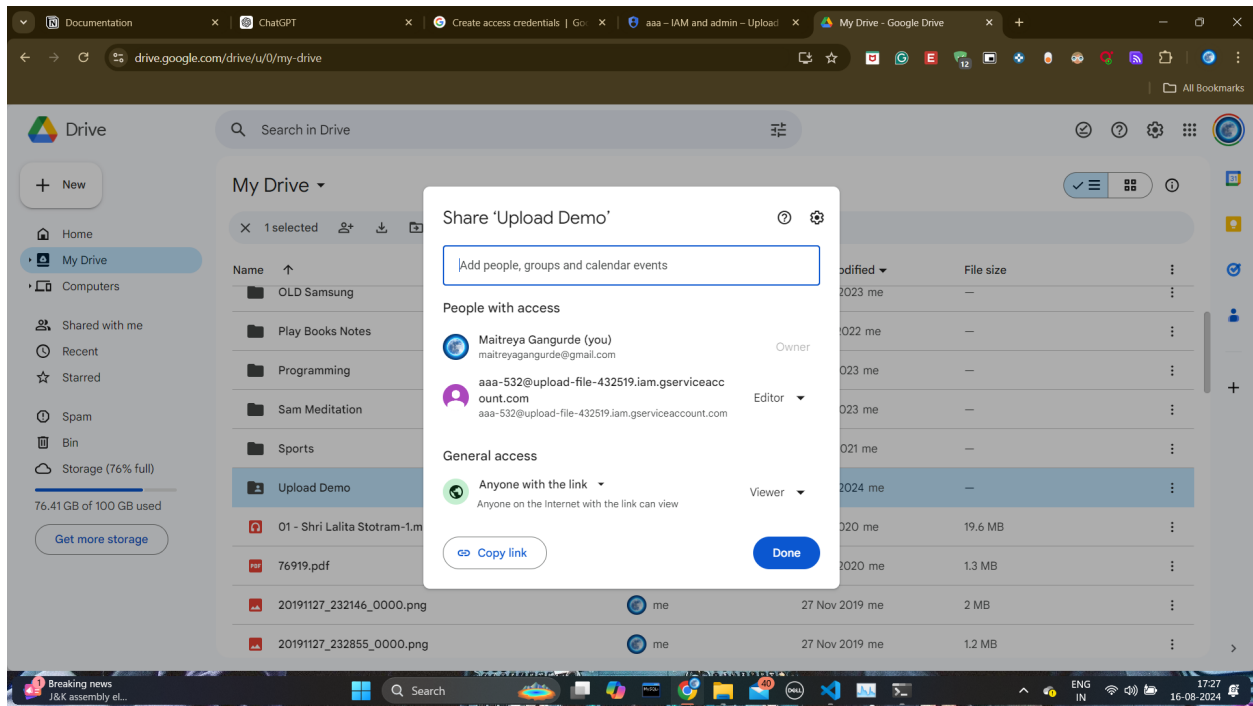
7. Select Manage keys



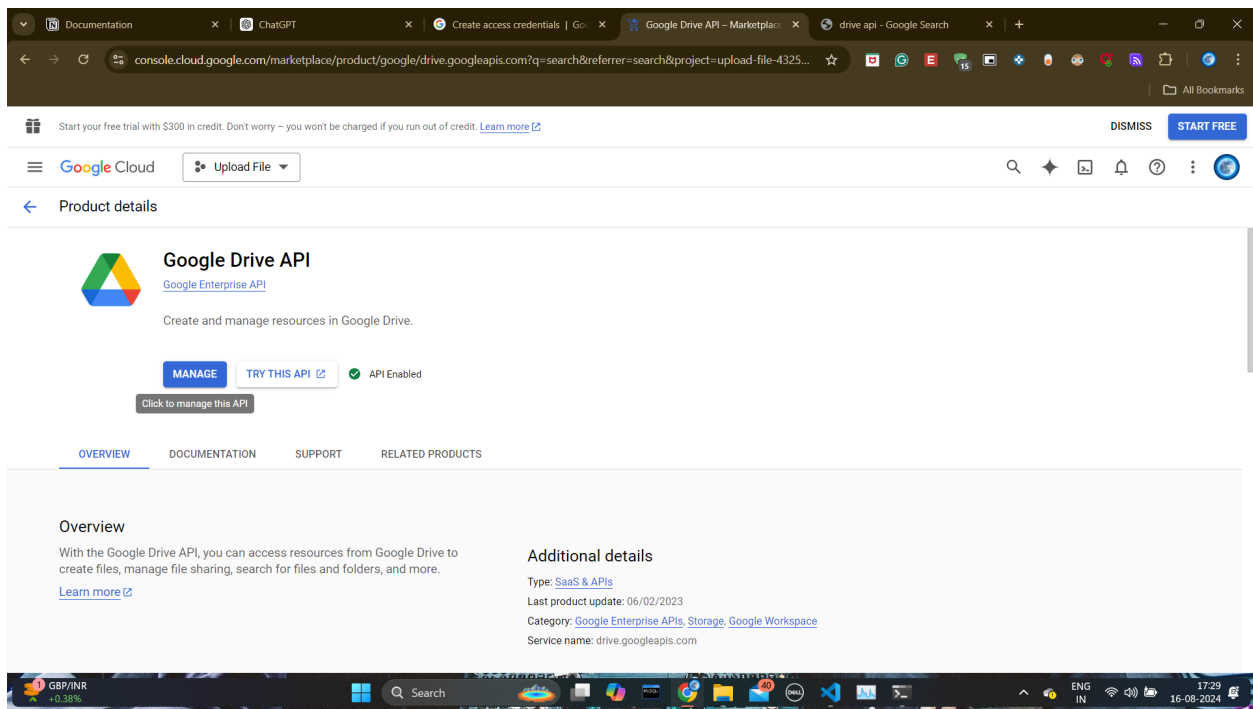
8. Create ADD key and select JSON

9. Then it will get automatically downloaded.

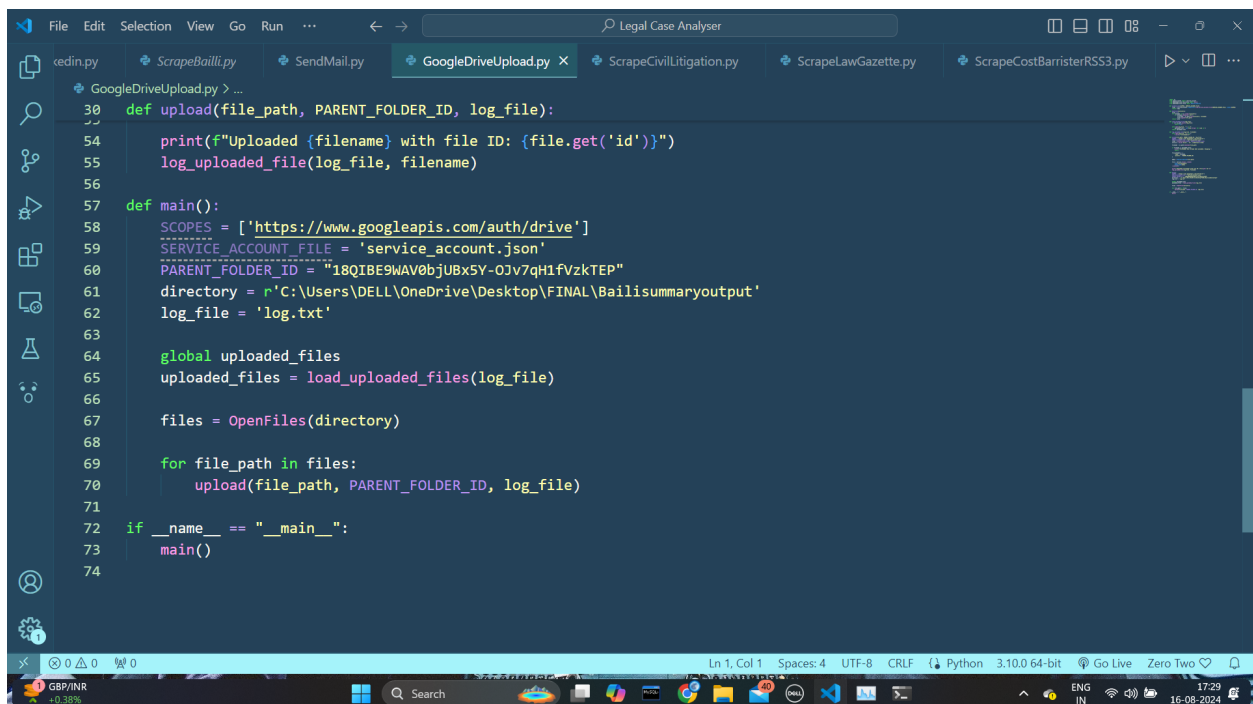
10. After downloading it copy that JSON file into our project path.
11. Copy the given email address.
12. Create a folder and into its manage access, make it open to add and give access to the email address which we copied.



13. Search drive API and enable it

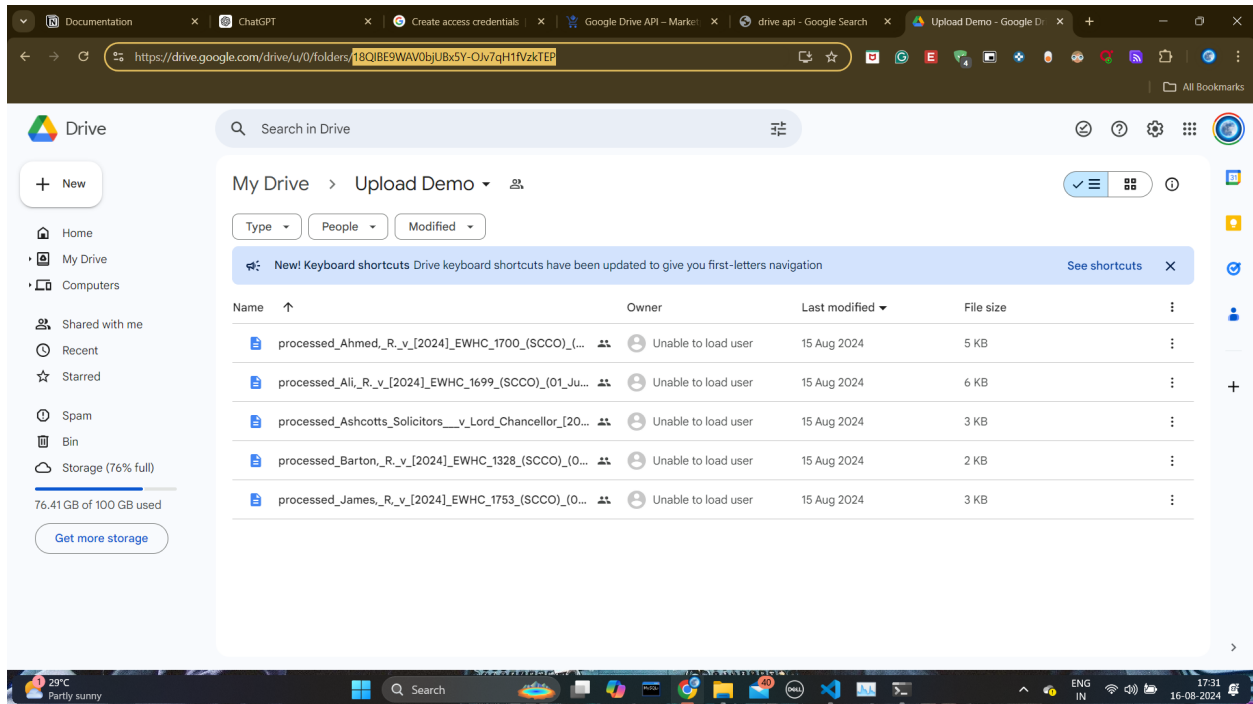


14. Now we have to make changes in the code accordingly.



see line no 60, there's a unique string to generate it we'll go and open the folder which we gave access.

15. Copy the selected string and paste it into the code.



Now all set run the script and then it will do its work.

Thank you :)